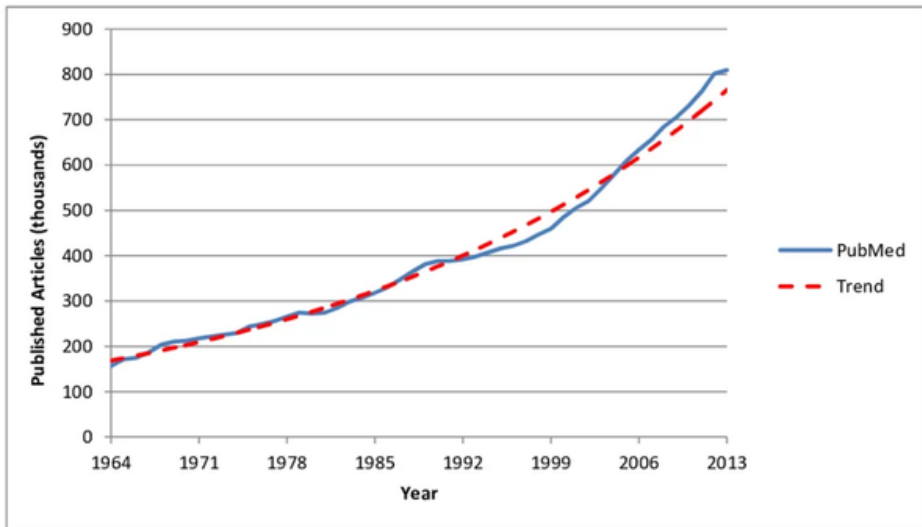


Problem & Motivation

Unlike general web searches, missing a single relevant document in a medical context can lead to incomplete literature reviews. Researchers often find themselves overwhelmed by information overload, where generic search tools return thousands of results, many of which lack the required depth or clinical relevance.



Malakasiotis et al. (2015). Biomedical question-focused multi-document summarization. BioASQ3.

Dataset

Medical document collection from Highwire Press, including TREC Genomics 2006-2007 to use for evaluation.

Metric Type	Count
Total Scientific Documents	162,259 Articles (49 Journals)
Corpus Lexicon	~994 Million Words
Preprocessed Tokens	1.06 Billion Tokens
Average Token Density	6542 tokens / document

ir-datasets.com/highwire.html

Technologies Used

- Python 3
- Pyterrier
- HuggingFace Transformers
- Streamlit Portal UI
- MeSH Vocabulary
- FastAPI + Uvicorn

Methodology

Preprocessing and Query Expansion (MeSH)

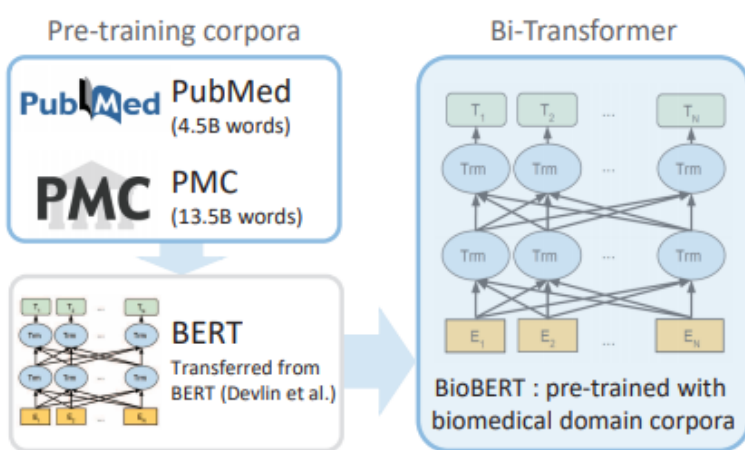
10 results for: *asthma*

Expanded query: asthma Asthmas Bronchial Anti Asthmatic Agents

Classical Document Retrieval (TF-IDF, BM25) and Boolean Retrieval

$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$
$$idf(t, D) = \log \left[\frac{N}{|\{d \in D : t \in d\}|} \right]$$
$$BM25 = \sum_{t \in q} \log \left[\frac{N}{df(t)} \right] \cdot \frac{(k_1 + 1) \cdot tf(t, d)}{k_1 \cdot \left[(1 - b) + b \cdot \frac{df(d)}{avg} \right] + tf(t, d)}$$

Neural Re-Ranking (BioBERT or PubMedBERT) (optional)



Displaying Results with Snippets on UI

10 results for: *nucleosome*

- > 1. 12637512 — Score: 14.6914 — Chromatin Remodeling Activities Act on UV-damaged Nucleosomes and
- > 2. 9593708 — Score: 14.6794 — Dynamics of Protein Kinase CK2 Association with Nucleosomes in Relatio
- > 3. 9826764 — Score: 14.6793 — Specific local histone-DNA sequence contacts facilitate high-affinity, non-

Methods Used

- Text Preprocessing** (tokenization, stopwords removal, stemming, etc.)
- Query Expansion** using Medical Subject Headings (MeSH) vocabulary
- Positional Inverted Index** (Phrase and proximity matching)
- Parameter tuning for BM25** (k1 and b parameters)
- Field Indexing** (Title and Journal search)
- Neural Reranking** (BioBERT / PubMedBERT)
- Pseudo Relevance Feedback** (Bo1 and KL models)

Evaluation and Results

Best k1 and b values for BM25:

Config	k1	b	MAP	P@5	NDCG@10	MRR
High Saturation	2.0	0.25	0.0716	0.1688	0.1655	0.3104

Top performance metrics evaluated using Highwire collection qrels:

Rank	Model Configuration	MAP	NDCG@10	P@5	MRR	Recall@10	Execution Time (seconds)
1	TF-IDF + KL	0.1260	0.2706	0.2656	0.4392	0.1160	32.6
2	TF-IDF + Bo1	0.1222	0.2567	0.2750	0.4170	0.1056	28.09
3	TF-IDF (Baseline)	0.1138	0.2437	0.2656	0.4141	0.1075	13.95
4	BM25 (k1=2.0, b=0.25) + KL	0.0910	0.1857	0.1906	0.3309	0.0866	32.59
5	BM25 (k1=1.5, b=0.00) + KL	0.0902	0.1806	0.1688	0.3101	0.0850	31.71

Discussion

- TF-IDF vs. BM25:** Surprisingly, TF-IDF outperformed BM25 across all combinations. Files in the Highwire corpus have the same structure, and since the length difference between the documents is not significant, the b parameter is not effective. The k1 parameter saturates medical terms, meaning that even if they appear more frequently after a certain point, they do not increase the score.
- Negative Effect of MeSH:** The users of this system already have a sufficient level of knowledge to write good queries. MeSH causes topic drift.
- Negative Effect of Neural Rerankings:** BERT models are limited to 512 tokens, which is not useful for the average 6542 tokens per document.
- Effect of PRF:** Bo1 and KL significantly boost MAP scores, with the trade-off from execution time.